Synthesis, Frontier Questions, And Student Research Designs

Turning Technology Questions Into Labor Research Designs

Special Topic 5: Technology, Innovation, and Labor – Week 6

Central question

- What makes a technology project a labor-economics paper?
- Start with the labor object: wages, employment, task content, mobility, training, job quality, bargaining power, or welfare.
- Then name the technology margin, adjustment channel, comparison, unit, timing, and incidence object.
- Week 6 goal: leave with project designs, not a generic future-of-work recap.

Course synthesis

Week	Design object
1. Tasks and frameworks	map technologies into tasks, skills, and equilibrium adjustment
2. Demand	distinguish displacement, augmentation, scale, and new tasks
3. Workers	study reskilling, mobility, career risk, and welfare costs
4. Firms	separate adoption, use, hiring, monitoring, and organization
5. Markets	trace rents, concentration, institutions, and distribution
6. Capstone	convert the sequence into credible research designs

Technology-to-labor research architecture

Design object	Required choice
Technology margin	invention, adoption, use, exposure, diffusion, implementation
Labor outcome	wages, employment, vacancies, training, mobility, job quality, welfare
Worker channel	retraining, task switching, occupation moves, separations, retirement
Firm channel	substitution, augmentation, scale, hiring, supervision, job redesign
Market incidence	local markets, sectors, platforms, concentration, institutions
Welfare object	rents, risk, autonomy, bargaining, adjustment costs, distribution

- A paper that names a technology has not yet named a labor mechanism.

Where the literature is

- Task frameworks and automation theory are mature: tasks, substitution, augmentation, and new work.
- Robot and local-exposure designs are the benchmark for market-level incidence.
- Worker-adjustment work measures earnings, mobility, training, and career risk better than subjective welfare or information frictions.
- Firm-adoption work now separates investment, adoption, use, skill demand, organization, and attitudes.
- Inequality work connects technology to rents, labor share, concentration, institutions, and worker voice.

What remains poorly measured

- Actual worker use rather than access, exposure, or adoption announcements.
- Organizational implementation after adoption.
- Incumbent training versus firms hiring around incumbents.
- Algorithmic management, monitoring intensity, autonomy, and voice.
- Job quality, schedule risk, promotion ladders, and bargaining power.
- Compute, electricity, and data-center constraints as labor-market forces.

Frontier directions

Frontier	Labor question
Generative AI use	who gets augmented, displaced, trained, or monitored?
Firm reorganization	how do vacancies, teams, supervision, and promotion change?
Compute infrastructure	which places and workers gain from AI buildout and bottlenecks?
Global service trade	which tasks become tradable, automated, or newly valuable?
Measurement	which technology proxy predicts which labor outcome?
Worker voice	how do platform rules and algorithmic governance change bargaining?

Research design choices

- Labor object: what outcome matters, and why is it a labor outcome?
- Technology margin: invention, availability, adoption, use, exposure, or implementation?
- Unit: worker, occupation, firm, establishment, team, platform, place, or country?
- Counterfactual: less exposed units, later adopters, randomized rollout, policy shift, or pre-trends?
- Timing: short-run productivity, medium-run hiring, long-run mobility, and equilibrium incidence can differ.
- Welfare: wages and employment may miss risk, autonomy, voice, rents, and mobility costs.

Common failure modes

- Technology-first framing with no labor mechanism.
- Treating a proxy as the technology itself.
- Confusing exposure with adoption or use.
- Ignoring selected adoption by firms, managers, or workers.
- Reading equilibrium incidence from partial-equilibrium evidence.
- Compressing timing into one coefficient.
- Interpreting wage effects as complete welfare effects.

How to find a student project

- Avoid saturated broad questions; enter through a sharper mechanism.
- Replace "Does AI matter?" with "Which labor outcome, for which unit, through which adjustment channel?"
- Look for measurement gaps: use versus exposure, organization versus adoption, worker voice versus productivity.
- Use setting-specific evidence only when it reveals a portable labor mechanism.
- Make the contribution legible to labor economists who do not study the specific technology.

Research Lab design

- Reproduce: build a synthetic local technology-exposure design inspired by robot-incidence research.
- Diagnose: classify what the measure captures and what it misses: adoption, use, exposure, timing, unit, welfare.
- Transfer: redesign the logic for generative AI worker use and firm adoption.
- Output: local exposure table, reduced-form incidence table, design diagnosis, and frontier transfer memo tables.
- Goal: practice turning technology measures into credible labor questions.

Takeaways

- Technology-and-labor research begins with work, workers, firms, markets, institutions, and welfare.
- The technology measure must match the mechanism it claims to identify.
- Strong designs state unit, timing, comparison, adjustment channels, and welfare limits.
- Frontier opportunities are practical: better measurement, worker use, firm organization, infrastructure, global services, and voice.
- The course ends when students can turn a fast-moving topic into a researchable labor question.